SKILLS
EXECUTIVE FUNCTION
CREATIVITY
COMMUNICATION
COLLABORATION
RESPONSIBILITY

ACTIVITY

▼ THE CHARACTERISTICS OF PERIPHERAL DEVICES

Create posters that include the characteristics and uses of a range of peripheral devices. You could split the work between members of your class so that some of you investigate input peripherals, some investigate output peripherals and some investigate storage peripherals.

INPUT

Input peripherals are devices that send data to the computer. They allow the user to control the computer or store data captured from sources outside the system.

KEYBOARDS

Keyboards use buttons known as keys, which users press to input text or to interact with software. Keyboards send data to the computer either using a wired or wireless connection. Different types of keyboards are used for different operating systems, languages and functions, as shown in Figure 1.35.



▲ **Figure 1.35** Different layouts are provided for different operating systems or language requirements

SUBJECT VOCABULARY

shortcut a combination of key presses used to access common software commands quickly

Combinations of key presses allow the user to access common software commands called **shortcuts**. Some people learn to touch-type, which is a method of typing without needing to look at the keys. This increases the speed at which these people can input data.



SUBJECT VOCABULARY

mapped linked

▲ **Figure 1.36** This is an example of a keyboard used with specialist video software; the keys are **mapped** to specialist commands

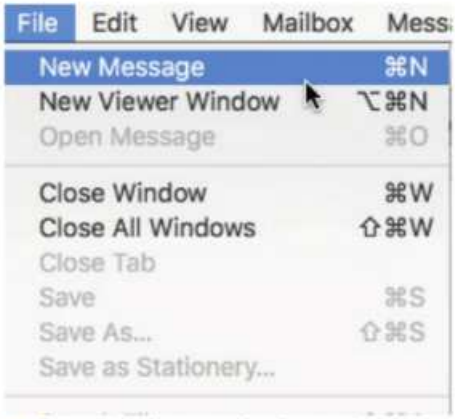
SKILLS PROBLEM SOLVING

DID YOU KNOW?

The layout of keys on a standard QWERTY keyboard comes from the layout of keys on a typewriter. The most commonly used letters were spread out, to reduce the number of times the metal typewriter hammers hit each other.

SUBJECT VOCABULARY

pointer an on-screen indicator used to select displayed objects



▲ **Figure 1.37** An arrow is often used as the pointer

ACTIVITY

▼ KEYBOARD SHORTCUTS

Choose a piece of software and look up three keyboard shortcuts that you could use to complete common tasks more quickly.

POINTING DEVICES

A **pointer** is used on a screen to select displayed objects. There are several types of device that allow you to control a pointer.

- **Mouse:** A mouse uses an optical sensor to recognise the movement of the device. Its sensitivity can be affected by the surface upon which it is placed. Some mice use roller balls to control the movement of the pointer.



▲ **Figure 1.38** Have you used a roller ball before?

- **Tracker ball:** A tracker ball is rolled to move the pointer. It does not have buttons to press.



▲ **Figure 1.39** Tracker balls are often used in installations like the kiosk shown here

- **Track pad:** The track pad surface senses finger movements, touches and presses.



▲ **Figure 1.40** Track pads are often built into laptop computers to save space

SUBJECT VOCABULARY

sprites on-screen graphics in a computer game, used to represent characters, vehicles and objects

- **Joystick:** Joysticks are commonly used on games controllers to move **sprites** around the game.



▲ **Figure 1.41** Have you used a joystick like this one?

SUBJECT VOCABULARY

stylus a pen-shaped device

- **Graphics tablet:** A graphics tablet is a flat pad that is used with a **stylus**. Users use the stylus to draw or write on the graphics tablet. These devices are often used by digital artists and designers.



▲ **Figure 1.42** Graphics tablets are often used by digital artists and designers



▲ **Figure 1.43** An OMR multiple choice response sheet

GENERAL VOCABULARY

characters printed or written letters and symbols

SCANNER (INCLUDING OCR AND OMR)

Scanners use light sensors to record physical documents as images, which are then saved as files to the computer.

Software allows scanners to read **characters** on the document and store the result in a text file. This is known as Optical Character Recognition (OCR).

Optical Mark Recognition (OMR) software can also be used with scanners to detect simple marks on a document. A common use for OMR is for recognising and recording responses to multiple choice tests.

SUBJECT VOCABULARY

barcode a pattern of lines and gaps that can be read by a device

BARCODE SCANNER

A **barcode** is a pattern of lines and gaps that can be read by barcode scanners, which detect the width of lines and gaps in a barcode. Barcodes are often used on parcels, so that they can be tracked, and on items for sale in shops. The barcode represents letters and numbers which are used to identify the item.

There are two types of barcode:

- linear
- matrix.



▶ **Figure 1.44** Linear barcode used to identify a parcel

Matrix codes are also known as QR codes. They are newer than linear barcodes and have some advantages over linear barcodes, such as:

- they can hold much more information than linear barcodes
- they can be scanned from any angle.

In South Korea, a supermarket company wanted to open a new shop, but there was no space left on the high street. Instead, the company created a 'virtual store' by putting posters of their supermarket shelves in a subway station, as shown in Figure 1.45. Every item pictured on the shelves had its own QR code that shoppers could scan. This allowed people to shop while they were travelling to work in the morning, knowing that their shopping would be delivered to their home that evening.

DID YOU KNOW?

There are apps for smartphones and tablet devices that use the device's camera to read barcodes and QR codes. There are also websites that allow users to create barcodes and QR codes.



▲ **Figure 1.45** A Virtual Subway Store in South Korea

SKILLS EXECUTIVE FUNCTION
RESPONSIBILITY
CO-OPERATION

SUBJECT VOCABULARY

plain text text that is not specially formatted

ACTIVITY

▼ MATRIX CODES

Visit a website to create a QR code for a **plain text** message. Ask a friend to scan it to reveal the message you have left for them.

WEBCAM

Webcams are specialised cameras. They are generally lower quality than camcorders, and may have built-in microphones to capture sound. They can be used as security cameras and can stream images or video to the internet.

SKILLS CRITICAL THINKING
INTELLECTUAL INTEREST AND CURIOSITY

ACTIVITY

▼ USING WEBCAMS

Webcam footage can be used for a variety of purposes. For example, if you do an internet search for 'Bondi Beach webcam', you will be able to see live footage from Bondi Beach in Australia, which can be used to check the conditions for surfing at the beach.

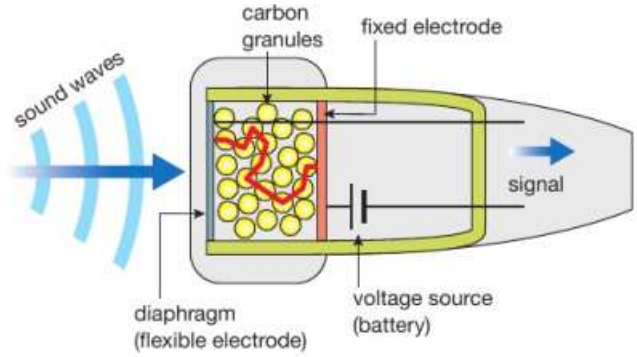
Find out about any webcams in your local country or local area. What are they being used for? Who would be interested in the footage?

SUBJECT VOCABULARY

diaphragm a thin round object, especially in a telephone or loudspeaker, that is moved by sound or that moves when it produces sound
analogue representing information with continuously variable electrical signals that digital signals approximate
sound card a device built into or added into a computer to allow it to playback and record audio
VoIP Voice over Internet Protocol, which is the technology that allows people to use broadband internet connections to speak to other people, rather than using a conventional telephone

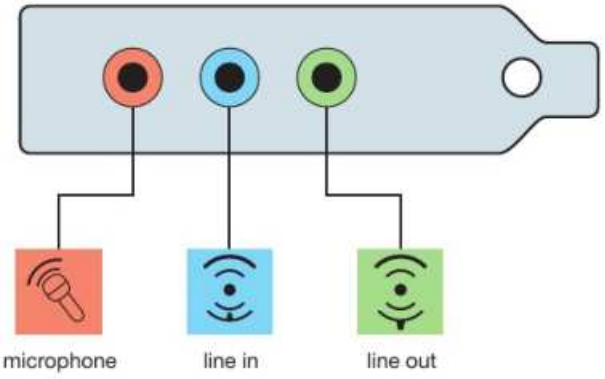
MICROPHONE

A microphone is used to capture sound. It uses a **diaphragm** which moves when air hits it. This movement produces an **analogue** electrical signal. A microphone converts the electrical signal to a digital signal with its **sound card**, which acts as an analogue-to-digital converter.



▲ **Figure 1.46** Sound waves hitting a microphone and being converted to an electrical signal

Small, low-quality microphones are often built into computers and are used for voice recognition, recording speech or allowing **VoIP** calls using software such as Skype©. Higher quality microphones are often used by recording artists and can be connected to a sound card's audio input port.



▲ **Figure 1.47** A microphone can be plugged into a port on the sound card



▲ **Figure 1.48** This headset contains a microphone and headphone speakers

DID YOU KNOW?

Microphones are often used in order to make devices and technology accessible to people with disabilities, as they provide access for people who are unable to input data using a mouse and keyboard.

HINT

Microphones can often be included in other devices, such as the headset in Figure 1.48. The headset itself is not a type of peripheral device, but a piece of equipment for combining a microphone and headphone speakers. Try to provide specific examples of peripheral devices when answering questions in the exam.

TOUCH SCREEN

Touch screens are used in many devices, including smartphones, tablet devices, laptops and desktop computers. There are two types of touch screen:

- resistive
- capacitive.

HINT

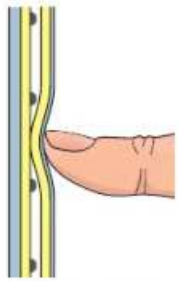
Do not confuse accessibility (increasing people's ability to access devices and technologies) with usability (making devices more efficient and satisfying to use).

GENERAL VOCABULARY

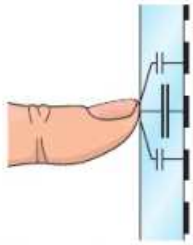
capacitive able to hold an electrical charge
conductive able to conduct electricity

When a user presses on a **resistive** screen, the pressure causes two layers underneath the screen to touch and make a connection. Resistive touch screens are more durable than capacitive touch screens, but are harder to read because more layers reflect more light. In addition, they can only recognise one touch at a time, so they are not suitable for multi-touch applications.

Under the glass of a **capacitive** screen, there is a layer of **capacitive** material. When a user touches the screen, a small amount of charge flows away from their finger because humans are **conductive**. The change in electrical charge is measured precisely: the closer the finger is to the charged areas, the more current flows away. This allows the computer to calculate the precise location at which the screen was touched. Capacitive touch screens are often used in smartphones.



▲ **Figure 1.49** A resistive touch screen



▲ **Figure 1.50** A capacitive touch screen



▲ **Figure 1.51** A fingerprint scanner detects unique patterns in your fingerprints

BIOMETRIC SCANNER

If you have watched a spy film, you may have seen a biometric scanner being used. Biometric scanners work by measuring part of the unique physical characteristics of a user, often multiple times, until a suitable average result is produced. This average result is then stored and future samples are compared against it in order to check whether the person being scanned is the authorised user. This allows the information to be updated so that the saved sample is perfected.

Four examples of biometric scanners are as follows.

- **Fingerprint recognition:** These scanners read the patterns of arches, loops and whorls in a human fingerprint. Fingerprints are unique to each individual person, which makes them a useful method of identification, but fingerprints can be obscured, damaged or changed, such as by injury or disease.



▲ **Figure 1.52** Examples of different types of fingerprint patterns

- **Facial recognition:** These scanners identify the structure of a human face in order to identify an individual.



▲ **Figure 1.53** Surveillance scanners are also used by some government agencies to recognise number plates, models and colours of vehicles

- **Voice recognition:** These scanners require a microphone to capture the voice. They then compare the voice print against a saved original and check to see whether the two prints match. (See page 26 for more information about microphones.)



▲ **Figure 1.54** Voice recognition software compares a voice print with a stored original and checks for a match

GENERAL VOCABULARY

iris the round, coloured part of an eye

- **Iris recognition:** Like fingerprints, the human iris has a unique pattern, though an iris scan is approximately 120 times more detailed than a fingerprint. This means that iris recognition is 120 times more reliable when identifying people.





▲ **Figure 1.55** Some passports now contain a chip that holds biometric data about the passport holder that is compared with the results of biometric scans carried out at airports

▼ **Table 1.2** Advantages and disadvantages of using biometric scanners

ADVANTAGES	DISADVANTAGES
Easier for users than passwords because biometric data cannot be forgotten	Cause privacy and security concerns because detailed personal information is stored electronically
More difficult to trick or confuse biometric scanners than some other methods	Possible to trick or confuse them
Can speed up transactions	Expensive to make and buy
Are not restricted by language barriers	Make some people feel uncomfortable

SKILLS
CRITICAL THINKING
COMMUNICATION
INTERPRETATION
ADAPTIVE LEARNING
EXECUTIVE FUNCTION
PERSONAL AND SOCIAL RESPONSIBILITY
INITIATIVE
COMMUNICATION
COLLABORATION
CO-OPERATION
INTERPERSONAL SKILLS
RESPONSIBILITY

ACTIVITY

▼ USING BIOMETRIC SCANNERS

1 Use the internet to do some research and create a table that compares the advantages and disadvantages of each method of biometric scanning (see *Unit 6 Software skills* (page 308) for more information about creating tables).

2 Create a survey to ask other people how they would feel about using each type of biometric scanner. What are their concerns?

3 What affects the reliability of each type of biometric scanner? Discuss your research findings and your thoughts with your teacher and the rest of your class.

4 What other factors could be considered when choosing a biometric security system?



▲ **Figure 1.56** Magnetic strip reader being used to unlock a door

CARD READERS

Card readers are used to read data stored on a card that is carried by a user. The data can be used to unlock doors, access secure areas, make payments or track people, parcels and even pet animals.

Cards can carry data using three methods. Each method needs a specialist type of reader.

- **Magnetic strip:** This is the least secure method as the data on the magnetic strip can easily be stolen by criminals who put the card through a card reader without the card owner's knowledge.
- **Programmable chip:** Data on a programmable chip is only readable when a correct PIN is entered into the reader. In 2005, the UK introduced the chip and PIN system, which reduced certain types of **fraud** by 67%.

SKILLS
PERSONAL AND SOCIAL RESPONSIBILITY
INTERPERSONAL SKILLS

GENERAL VOCABULARY

fraud deceiving other people for criminal reasons

ACTIVITY

▼ FRAUD

In the UK, online fraud rose 79% in the first three years after the country switched to the chip and PIN system. Discuss why this may have happened.



▲ **Figure 1.57** Examples of cards using chip and PIN

- **Radio Frequency Identification (RFID) and Near Field Communication (NFC):** RFID is a short-range wireless communication method, and Near Field Communication (NFC) is a branch of RFID. RFID tags are cheap and small and they can be included in a variety of objects such as cards, key fobs and smartphones. They contain a unique identification number (ID) that is linked to records in a database. For example, the first image in Figure 1.58 shows a traveller using a prepaid travel card. A unique ID stored on the card links to account information stored in a database. You will learn more about databases in Unit 6 Software skills (page 307).



▲ **Figure 1.58** RFID tags can be included in a variety of objects

SENSORS

Sensors are used to input data about the physical environment. They can automatically input data without the need for human action. Their output is then processed by a computer. Sensors are one of the main features of smart devices such as smartphones and smart environments such as **smart homes**.

SUBJECT VOCABULARY

sensor an input peripheral that inputs data about the physical environment
smart home a home equipped with devices that can connect to the internet and be controlled remotely by a computer or smartphone app



▲ **Figure 1.59** (Left to right) motion sensor, **proximity** sensor, temperature sensor and button sensor; the red item at the top is a light sensor and the red item at the bottom is a UV sensor

GENERAL VOCABULARY

proximity how close something is relative to something else

The use of sensors has many benefits, such as:

- they can be placed in remote or dangerous places
- they can monitor continually
- they remove the possibility of human error
- they can sense things that people cannot (such as small changes in pressure or gas levels)
- the data is easily (sometimes automatically) converted to a digital form.

There are many different types of sensor to allow for the monitoring of a range of environmental factors. Some of these factors are:

- light
- moisture and humidity
- temperature
- proximity and distance
- motion and movement (such as Passive Infrared (PIR) sensors)
- pressure.

SKILLS CRITICAL THINKING
INTELLECTUAL INTEREST AND CURIOSITY

ACTIVITY

▼ USING SENSORS

1 For each environmental factor listed above, state one remote and one dangerous environment where a sensor could be used.

2 Research the appliances in modern smart homes that can use sensors. For each one, describe how sensors are used.

3 Research how sensors can help to prevent air traffic accidents. Can you find any instances where they may have caused accidents?



▲ **Figure 1.60** Sensors are used in aircraft and data from them are displayed to the flight crew on instruments in the cockpit

OUTPUT

Output peripherals are connected to a computer and output the results of the computer's processing in various forms, including:

- electronic display
- printed text
- video
- audio
- tactile (touch) forms.

MONITOR

Monitors allow users to see the output from the computer on an electronic display. Features of monitors include:

- screen size, which is measured diagonally
- resolution, which is measured in pixels (see pages 9–10 for information about pixels and screen resolutions)
- energy efficiency measures, such as going into a low-power standby mode if no input is detected.

Some monitors also use touch screen technology (see page pages 26–27 for more information about touch screens).



▲ **Figure 1.61** Monitors have evolved over time to provide better quality and more features

SKILLS COMMUNICATION EXECUTIVE FUNCTION

ACTIVITY

▼ USING MONITORS

1 Discuss the different uses for monitors.

2 Create a table of the uses for monitors with different sizes and resolutions (see *Unit 6 Software skills* (pages 220–223) for more information about creating tables).

PRINTER

There are three types of standard printer: dot matrix, inkjet and laser.



▲ **Figure 1.62** Laser (top left), inkjet (right) and dot matrix (bottom left) printers

GENERAL VOCABULARY

carbon copy a copy of a document made using carbon paper (paper coated in a coloured substance that leaves traces on clean paper)

- **Dot matrix:** These printers are sometimes called 'impact' printers. This is because the print is made by hitting or 'impacting' the paper through a ribbon of ink, and this process is very noisy. The paper often has a number of **carbon copy** layers so it can make several copies of the same document as it prints it.



▲ **Figure 1.63** A dot matrix printer

- **Inkjet printers:** These printers use cartridges containing different coloured ink. Black ink is held in a one cartridge, while the colour cartridge is often split into three colours: cyan, magenta and yellow. Tiny dots of these inks are sprayed onto the paper to form images or text.

DID YOU KNOW?

Many cartridges can be refilled to save money and protect the environment.



▲ **Figure 1.64** Cartridges used in an inkjet printer

SUBJECT VOCABULARY

discharge remove the electrical charge from something
toner a type of ink

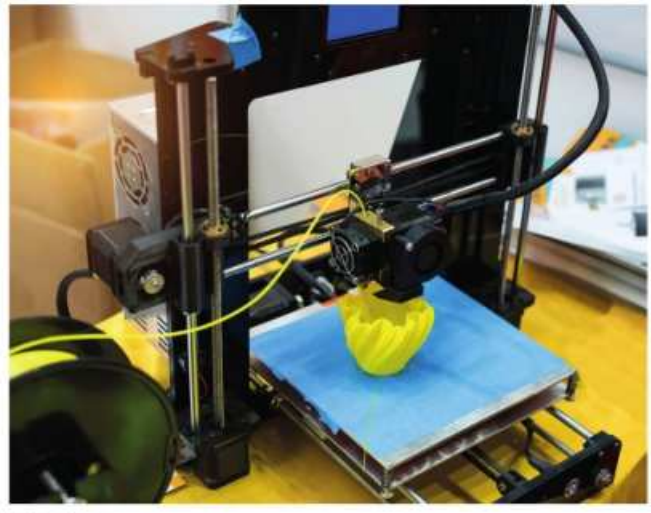
- **Laser printers:** These printers contain a rotating cylinder or drum that holds an electrical charge. A laser is used to **discharge** certain points on the drum and 'draw' an image. Electrically charged **toner** is attracted to those points on the drum and is heated onto paper that is passed over the drum.

▼ Table 1.3 Comparing printer types

	SPEED	PRINTER COST	COST PER COPY	COLOUR PRINTING	QUALITY	EXAMPLES OF WHERE IT IS USED
Dot matrix	Slowest	Cheap	Low	Rare	Low	When multiple copies are required.
Inkjet	Fast	Medium	High	Yes	High	Home printing and photographs
Laser	Fastest	Expensive	Medium	Yes	Highest	High-volume printing, often used in offices

3D PRINTER

Three-dimensional (3D) printers create real-life versions of digital models. These printers work by adding layers of heated material from the bottom of the model all the way to the top. 3D-printed models take a long time to create.



▲ Figure 1.65 A 3D printer

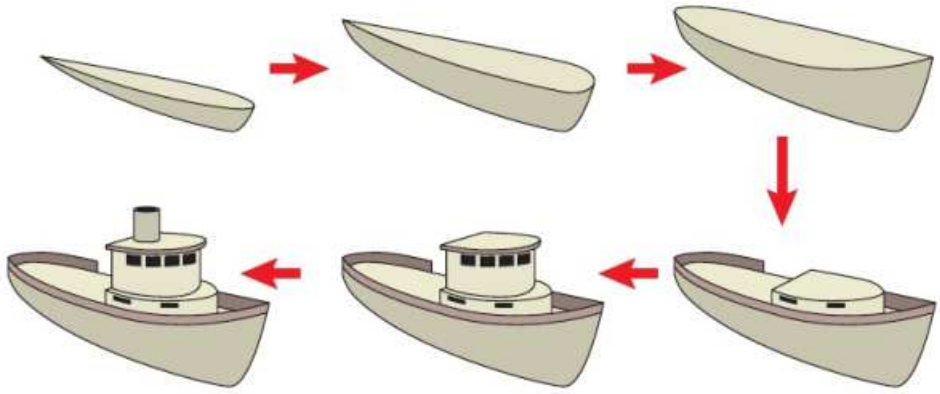
SKILLS INTELLECTUAL INTEREST AND CURIOSITY

ACTIVITY

▼ USING 3D PRINTERS

Find a video online that shows a 3D printer in action.

3D printers are often used to create complex items, sometimes with working parts. Some of these items could not be assembled in any other way. Examples of 3D-printed items include aeroplane parts, artificial limbs, prototype models of cars and even food such as pasta and chocolate.



▲ **Figure 1.66** Stages of building a 3D-printed model

SKILLS INTELLECTUAL INTEREST AND CURIOSITY
PERSONAL AND SOCIAL RESPONSIBILITY
ETHICS
COMMUNICATION
REASONING/ARGUMENTATION

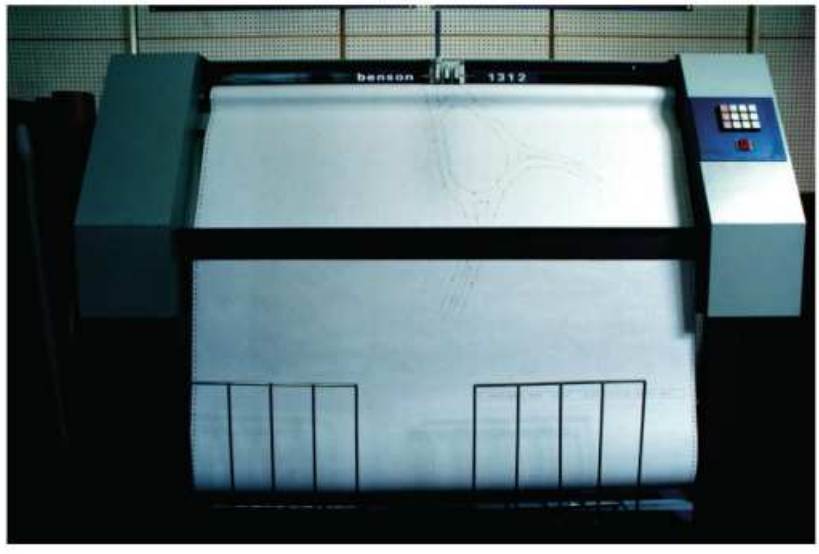
ACTIVITY

▼ TECHNOLOGY AND ETHICS

1 In 2013, someone published the plans for a 3D-printed gun online. Discuss the impact of this.

2 Write notes to support two opposing arguments. One argument should support the wide availability of 3D printers, and the other should argue against the wide availability of these printers. You could then use your notes to debate this topic in your class.

PLOTTER



▲ **Figure 1.67** A plotter

DID YOU KNOW?
Plotters can also use a cutter attachment instead of the pen attachment to cut vinyl. This can be used to create stickers or images to go on printed clothing.

GENERAL VOCABULARY
cartographers people who draw and produce maps

Plotters are used by a variety of professionals, including product designers, architects, engineers and **cartographers**. The plotter draws high-quality images created in Computer Aided Design (CAD) software onto large sheets of paper. It does this by moving a pen across the paper to draw the lines, and the pen can be raised away from and lowered onto the sheet of paper. Some plotters can also roll the paper backwards and forwards.

DATA PROJECTOR

Data projectors are used for home entertainment, such as watching television or playing games, and for giving presentations in classrooms and business meetings. They have the following features.

- **Light bulbs:** These bulbs can have various brightness levels (measured in lumens) and are expensive to replace.
- **Resolution:** Like monitors, they use different quality screens (see page 9–10 for more about screen resolutions).
- **Zoom functionality:** The projected image can be made larger by enlarging the image using either **optical zoom** or **digital zoom**.
- **Portability:** Some smaller and lighter projectors are available, and these are easier to move around.

SUBJECT VOCABULARY

optical zoom focusing a lens in order to zoom in on an object
digital zoom enlarging part of an image to produce the effect of zooming in



▲ **Figure 1.68** Data projectors are used to project a large image (or film) onto screens

SKILLS ADAPTABILITY SELF-DIRECTION

ACTIVITY

▼ TYPES OF PROJECTOR

1 Do some online research to find an appropriate projector for:
   - use in a classroom
   - a home entertainment system
   - a person travelling on business.
2 For each projector that you identify, find out how long its light bulbs last (on average) and how much the replacement bulbs cost.

SPEAKER

Speakers allow a computer to output sound. Speakers often come in pairs to provide **stereo sound**. Multiple speakers are commonly used to provide **surround sound** in home entertainment systems. These speakers need a special amplifier that can deliver different levels of sound to each speaker to provide **spatial awareness**.

SUBJECT VOCABULARY

stereo sound sound that comes from two sides (left and right)
surround sound sound that comes from four or more sides, used so that sounds from a film or television programme come from all around the viewer as they would in real life

GENERAL VOCABULARY

spatial awareness the impression of a realistic space



▲ **Figure 1.69** A pair of speakers is required for stereo sound because one is used for the left side and one is used for the right side

SKILLS SELF-DIRECTION PROBLEM SOLVING

SUBJECT VOCABULARY

mono sound sound that comes from a single speaker

ACTIVITY

▼ DIFFERENT TYPES OF SOUND

Do some research to find out when **mono sound** may be most appropriate. When might listeners prefer stereo sound?

A single speaker is usually used to provide low-quality audio feedback to a user. These speakers are often small and low-quality, and are often found in alert systems such as alarms and buzzers.

SKILLS ADADPTIVE LEARNING

ACTIVITY

▼ USING LOW-QUALITY SPEAKERS

Think of the appliances and devices in your home that use low-quality speakers (you might want to start in the kitchen!).



▲ **Figure 1.70** A piezoelectric 'piezo' speaker used in a watch

CONTROL DEVICE

Control devices are also known as actuators, and they are components of a system that make something happen in the real physical world. These devices are often mechanical. Examples of actuators include:

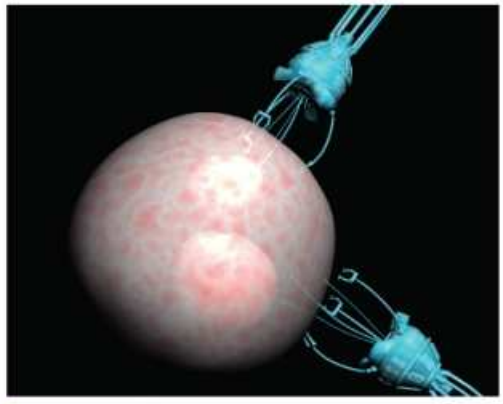
- valves
- pistons
- heaters
- coolers
- motors.



▲ **Figure 1.71** Robots use actuators that are powered in many ways such as electricity, chemicals and compressed air

DID YOU KNOW?

Nanorobots are made up of tiny sensors and actuators that are capable of assisting a surgeon by carrying out very precise medical procedures.



▲ **Figure 1.72** Nanorobots allow doctors to make smaller incisions or cuts and more precise movements than they could if they used their own hands